

KARNATAK LAW SOCIETY'S
GOGTE INSTITUTE OF TECHNOLOGY

UDYAMBAG, BELAGAVI – 590008

(An Autonomous Institution under Visvesvaraya Technological University,
Belagavi)

(APPROVED BY AICTE, NEW DELHI)



Major Project Proposal

submitted in the partial fulfillment for the academic requirement of
6th Semester BE in Information Science and Engineering

**TITLE: Ontology-Aware Protein Function Prediction
from Primary Sequence Using Frozen Protein
Language Models**

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Batch no.: 21

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CERTIFICATE

This is to certify that Ms. Khushi Patil, Ms. Diya Kalghatgi and Ms. Nidhi Nayak of 6th semester and bearing USN 2GI23IS039, 2GI23IS030 and 2GI23IS056 has satisfactorily completed the Major Project. It can be considered as a bonafide work carried out in partial fulfilment for the academic requirement of 6th Semester B.E. Information Science and Engineering prescribed by KLS Gogte Institute of Technology, Belagavi during the academic year 2025- 2026.

The report has been approved as it satisfies the academic requirements in respect of Major Project – Phase 0 prescribed for the said Degree.

Signature of the Faculty Member

Abstract

Protein function annotation is a fundamental task in computational biology, with significant applications in drug discovery, disease analysis, genomics, and biotechnology. However, the rapid growth of protein sequence databases has far exceeded the pace of experimental function validation, creating a critical need for accurate automated prediction systems. This project proposes an ontology-aware deep learning framework for predicting protein functions directly from amino acid sequences using pretrained ESM-2 transformer embeddings and a Multi-Layer Perceptron (MLP) classifier. The pretrained ESM-2 model is utilized as a frozen feature extractor to generate high-dimensional sequence representations, which are then used to train a multi-label classifier for Gene Ontology (GO) term prediction across Molecular Function, Biological Process, and Cellular Component categories. To improve biological consistency, a hierarchical loss mechanism is incorporated based on parent-child relationships within the Gene Ontology structure. The proposed system aims to deliver efficient and scalable protein annotation without requiring structural data, while reducing computational cost compared to full model fine-tuning. Performance will be evaluated using standard metrics such as Precision, Recall, F1-score, Fmax, and Area Under Precision-Recall Curve (AUPR). The final model will be deployed through a user-friendly web interface for practical accessibility. This work demonstrates the integration of transformer-based protein language models and ontology-guided learning for next-generation bioinformatics applications.

Literature Survey

Recent research in protein function prediction has increasingly focused on the application of deep learning and transformer-based language models due to their ability to learn rich biological representations directly from amino acid sequences. Traditional homology-based approaches often fail when similar annotated sequences are unavailable, motivating the development of data-driven methods using neural networks. Various studies have explored convolutional networks, recurrent architectures, graph neural networks, and pretrained protein language models such as ESM and ProtBERT to improve annotation accuracy and scalability. These works collectively emphasize the importance of sequence embedding extraction, multi-label classification, ontology-aware learning, and hierarchical consistency in enhancing Gene Ontology (GO) term prediction. Overall, transformer-based systems significantly outperform conventional similarity-based methods and demonstrate strong potential for large-scale automated protein annotation in modern bioinformatics applications [1]–[6].

Kulmanov et al. proposed **DeepGOPlus**, a CNN-based protein function prediction framework that combines sequence motif learning with sequence similarity search. Their model improved prediction accuracy while remaining computationally efficient, making it suitable for large-scale annotation tasks [1].

Glorigjević et al. introduced **DeepGraphGO**, a graph neural network-based approach that integrates protein interaction networks with sequence-derived features for function prediction. Their work demonstrated that graph-based representations significantly enhance the prediction of complex biological functions [2].

Rives et al. developed large-scale transformer protein language models trained on millions of protein sequences, showing that unsupervised learning captures structural and functional properties of proteins effectively. Their findings laid the foundation for models such as ESM used in downstream prediction tasks [3].

Elnaggar et al. presented **ProtTrans**, a family of transformer-based protein language models including ProtBERT and ProtT5. Their study showed that transfer learning from massive protein databases enables highly informative embeddings that improve multiple bioinformatics tasks, including function prediction [4].

Fang et al. proposed an ontology-aware deep learning framework that incorporates Gene Ontology hierarchical dependencies into the loss function for protein annotation. Their results showed that hierarchy-guided learning improves biological consistency and multi-label prediction performance [5].

You et al. conducted a comparative survey of artificial intelligence methods for automated protein function prediction, highlighting the superiority of transformer-based embeddings and hybrid neural architectures over conventional sequence alignment methods. The study also discussed challenges such as class imbalance, rare GO terms, and model interpretability [6].

Research Gap

Despite significant advancements in protein function prediction using deep learning and transformer-based protein language models, several limitations still exist in current research. Many existing models achieve strong performance but are often evaluated on limited or highly curated datasets, reducing their ability to generalize to diverse real-world protein sequences. A large number of approaches primarily focus on maximizing prediction accuracy while **insufficiently utilizing the hierarchical relationships present in Gene Ontology, leading to biologically inconsistent outputs where child terms may be predicted without corresponding parent terms**. Furthermore, several state-of-the-art transformer models require extensive computational resources for fine-tuning, making them less practical for institutions with limited hardware support. Another major challenge is the severe class imbalance in Gene Ontology annotations, where rare functional terms receive poor predictive performance. In addition, many current systems lack interpretability and user-friendly deployment platforms for practical usage by researchers. Therefore, there is a need for a computationally efficient, ontology-aware, scalable, and deployable protein function prediction framework that leverages pretrained embeddings while maintaining biological consistency and strong real-world applicability.

Problem Statement

A massive gap exists between the number of known protein sequences and experimentally verified protein functions, primarily due to the rapid growth of sequencing technologies. Manual annotation through laboratory experiments is slow, expensive, and resource-intensive, making it impractical for large-scale analysis. Existing computational methods often fail to capture deep sequence patterns, depend heavily on sequence similarity, or ignore the hierarchical structure of biological functions defined in Gene Ontology. As a result, many proteins remain partially annotated or completely uncharacterized, limiting progress in genomics, drug discovery, and disease research.

Therefore, there is a need for an automated, scalable, accurate, and biologically consistent protein function prediction system using advanced deep learning techniques that can efficiently analyze large volumes of sequence data.

Objectives

1. To develop an automated deep learning-based system for predicting protein functions directly from amino acid sequences using pretrained ESM-2 embeddings.
2. To design a multi-label classification model that incorporates Gene Ontology hierarchical relationships for accurate and biologically consistent function prediction.
3. To evaluate and deploy the proposed model through a user-friendly platform for scalable real-world protein annotation applications.

Proposed Methodology

Step 1: Data Collection

Protein sequences and annotations will be collected from:

- UniProt SwissProt Database
- Gene Ontology Consortium

Step 2: Data Preprocessing

- Remove duplicate sequences
- Filter noisy annotations
- Keep experimentally verified labels
- Convert labels into multi-hot encoded vectors

Step 3: Feature Extraction

Use pretrained ESM-2 transformer model to generate embeddings from protein sequences.

Step 4: Classification Model

Train a 3-layer MLP neural network on extracted embeddings for multi-label GO prediction.

Step 5: Hierarchical Consistency

Apply Gene Ontology hierarchy loss to maintain biologically valid parent-child predictions.

Step 6: Evaluation

Use:

- Precision
- Recall
- F1-score
- Fmax
- AUPR

Step 7: Deployment

Develop a Streamlit web application where users can input FASTA sequences and receive predicted functions.

Tools and Technologies

Category	Tools
Programming	Python
Deep Learning	PyTorch
Bioinformatics	Biopython, Goatools
ML Utilities	NumPy, Pandas, Scikit-learn
Visualization	Matplotlib, Seaborn
Deployment	Streamlit
Dataset	UniProt, GOA

Possible Outcomes

1. The proposed system is expected to provide accurate prediction of protein functions directly from amino acid sequence data using advanced deep learning techniques.
2. It will significantly reduce the time required for protein annotation compared to traditional manual and laboratory-based methods.
3. The project will deliver a scalable AI-driven framework capable of handling large biological datasets efficiently.
4. A user-friendly web-based interface will be developed to enable easy access for researchers, students, and bioinformatics professionals.

References

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Curated protein sequences with expert-reviewed annotations. Best dataset for your project.
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